

SQL Curriculum Outline

The curriculum was taught progressively using a word dictionary and, later, Scrabble data as a common dataset. Each new SQL concept was introduced as it became useful for solving a practical problem.

1. SQL Server Management Studio

- Connecting to SQL Server through SSMS
- Navigating databases and database objects

2. Querying and Filtering Data

- Basic queries
- Filtering records
- Understanding result sets

3. Schemas

- What a schema is
- Creating and using schemas
- Organizing database objects

4. Importing Data

- Importing flat-file data using the SQL Server import wizard
- Renaming columns during import
- Identifying row headers
- Handling file formatting
- Introduction to SQL Server Integration Services (SSIS)

5. Database Context

- Using the USE command
- Understanding the current database context

6. Creating and Populating Tables

- Creating tables
- Defining columns and datatypes
- INSERT INTO
- SELECT ... INTO
- Understanding the difference between inserting into an existing table and creating a table from query results

7. Scripting Database Objects

- Generating table scripts
- Reading and modifying generated SQL
- Recreating database objects from scripts

8. Primary Keys and Indexes

- Creating primary keys
- Why primary keys matter
- Introduction to indexes
- How indexes relate to data retrieval and table design

9. Views

- Creating views
- What a view is
- When and why views are useful
- SQL comments using `--` and `/* ... */`

10. Stored Procedures

- What stored procedures are
- Why reusable database logic is useful
- Introduction to procedural SQL

11. Adding Derived Columns

- Adding an ordered or sequential number to a result set or table
- Using generated values to support later analysis

12. Temporary Data Structures

- Local temporary tables
- Global temporary tables
- Table variables
- When each type is useful

13. Parameters and Variables

- Declaring variables
- Using parameters
- Passing values into reusable SQL logic

14. Guided Analytics Exercises

The dictionary dataset was then used to apply the preceding concepts through progressively more difficult analytical questions.

Basic Aggregation and Filtering

- How many words are in the dictionary?
- How many words begin with A?
- Which starting letter has the most words?
 - `ORDER BY`
 - `SELECT TOP`

- How many one-letter words are there?
- Which word length occurs most frequently?

Ordering and Ranking

- What is the 200th word?
- What is the 200th word from the end?
- What is the longest word?

Pattern Analysis

- How many words contain the letter X?
- How many words begin with X?
- How can we identify all palindromes?
- How can we identify anagrams?
 - Joins
 - Views

Scrabble Exercise

Homework: Determine the Scrabble value of every word.

- Create a table variable containing letter point values
- Import a Scrabble dictionary
- Calculate the score of each word
- Determine whether words from the primary dictionary are valid Scrabble words
- Combine multiple datasets to produce the final result

Additional Challenges

- Which word contains the most vowels?
- Which word contains the most consonants?

1.